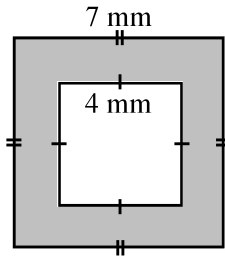


Composite Area - Subtraction

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Find the shaded areas of the following shapes

1



$$A_1 = l^2$$

$$A_2 = l^2$$

=

=

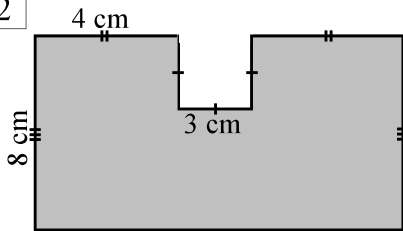
$$A_1 = \text{mm}^2$$

$$A_1 = \text{mm}^2$$

$$A = A_1 - A_2 = -$$

$$A =$$

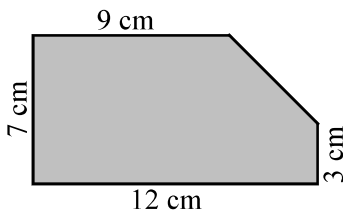
2



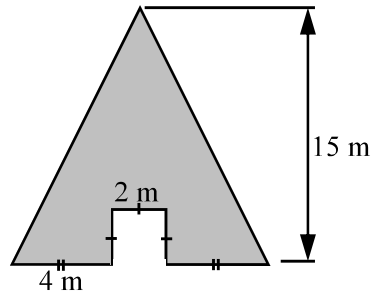
$$A = A_1 - A_2 = -$$

$$A =$$

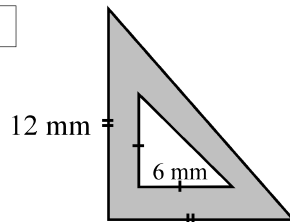
3



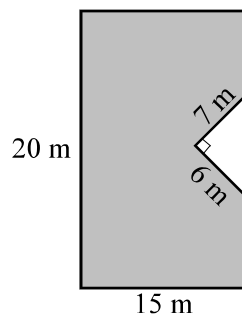
4



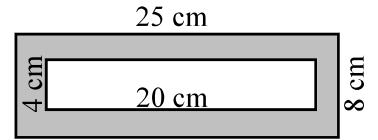
5



6

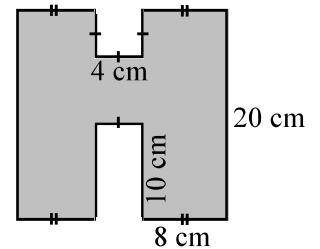


7



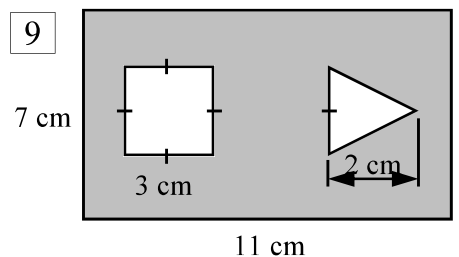
Now subtract two areas

8



$$A = A_1 - A_2 - A_3$$

9

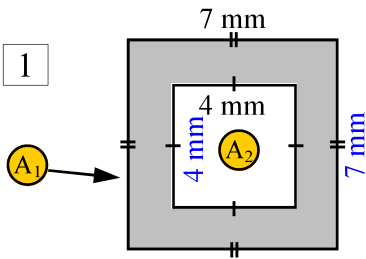


Composite Area - Subtraction

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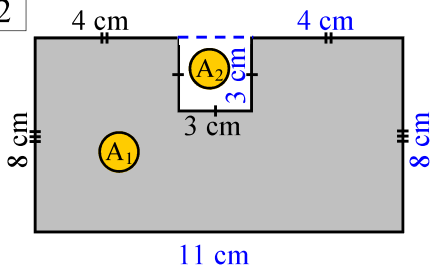
Find the shaded areas of the following shapes

1



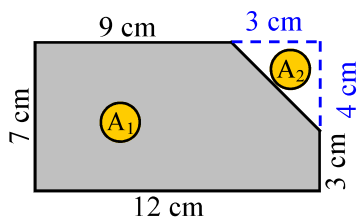
$$\begin{aligned} A_1 &= l^2 \\ &= 7^2 \\ A_1 &= 49 \text{ mm}^2 \\ A_2 &= l^2 \\ &= 4^2 \\ A_2 &= 16 \text{ mm}^2 \\ A &= A_1 - A_2 = 49 - 16 \\ A &= 33 \text{ mm}^2 \end{aligned}$$

2



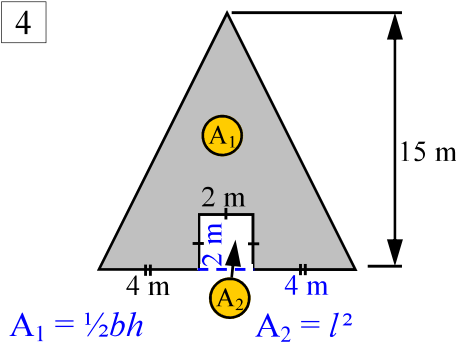
$$\begin{aligned} A_1 &= lb \\ &= 11 \times 8 \\ A_1 &= 88 \text{ cm}^2 \\ A_2 &= l^2 \\ &= 3^2 \\ A_2 &= 9 \text{ cm}^2 \\ A &= A_1 - A_2 = 88 - 9 \\ A &= 79 \text{ cm}^2 \end{aligned}$$

3



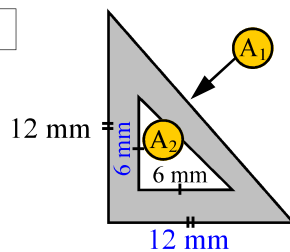
$$\begin{aligned} A_1 &= lb \\ &= 12 \times 7 \\ A_1 &= 84 \text{ cm}^2 \\ A_2 &= \frac{1}{2}bh \\ &= \frac{1}{2} \times 4 \times 3 \\ A_2 &= 6 \text{ cm}^2 \\ A &= A_1 - A_2 = 84 - 6 \\ A &= 78 \text{ cm}^2 \end{aligned}$$

4



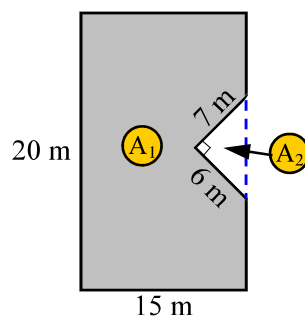
$$\begin{aligned} A_1 &= \frac{1}{2}bh \\ &= \frac{1}{2} \times 10 \times 15 \\ A_1 &= 75 \text{ m}^2 \\ A_2 &= l^2 \\ &= 2^2 \\ A_2 &= 4 \text{ m}^2 \\ A &= A_1 - A_2 = 75 - 4 \\ A &= 71 \text{ m}^2 \end{aligned}$$

5



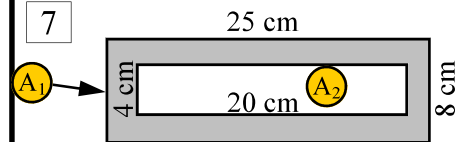
$$\begin{aligned} A_1 &= \frac{1}{2}bh \\ &= \frac{1}{2} \times 12 \times 12 \\ A_1 &= 72 \text{ mm}^2 \\ A_2 &= \frac{1}{2}bh \\ &= \frac{1}{2} \times 6 \times 6 \\ A_2 &= 18 \text{ mm}^2 \\ A &= A_1 - A_2 = 72 - 18 \\ A &= 54 \text{ mm}^2 \end{aligned}$$

6



$$\begin{aligned} A_1 &= lb \\ &= 15 \times 20 \\ A_1 &= 300 \text{ m}^2 \\ A_2 &= \frac{1}{2}bh \\ &= \frac{1}{2} \times 6 \times 7 \\ A_2 &= 21 \text{ m}^2 \\ A &= A_1 - A_2 = 300 - 21 \\ A &= 279 \text{ m}^2 \end{aligned}$$

7

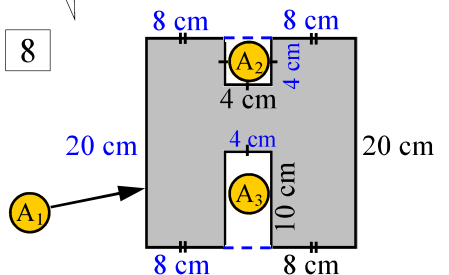


$$\begin{aligned} A_1 &= lb \\ &= 25 \times 8 \\ A_1 &= 200 \text{ cm}^2 \\ A_2 &= lb \\ &= 20 \times 4 \\ A_2 &= 80 \text{ cm}^2 \\ A &= A_1 - A_2 = 200 - 80 \\ A &= 120 \text{ cm}^2 \end{aligned}$$



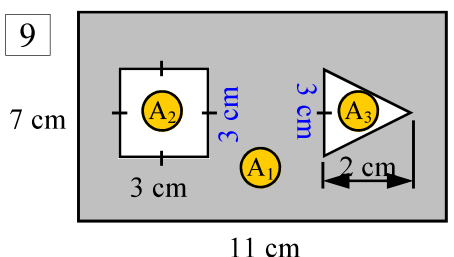
Now subtract two areas

8



$$\begin{aligned} A_1 &= l^2 \\ &= 20^2 \\ A_1 &= 400 \text{ cm}^2 \\ A_2 &= l^2 \\ &= 4^2 \\ A_2 &= 16 \text{ cm}^2 \\ A_3 &= lb \\ &= 4 \times 10 \\ A_3 &= 40 \text{ cm}^2 \\ A &= A_1 - A_2 - A_3 \\ &= 400 - 16 - 40 \\ A &= 344 \text{ cm}^2 \end{aligned}$$

9



$$\begin{aligned} A_1 &= lb \\ &= 11 \times 7 \\ A_1 &= 77 \text{ cm}^2 \\ A_2 &= l^2 \\ &= 3^2 \\ A_2 &= 9 \text{ cm}^2 \\ A_3 &= \frac{1}{2}bh \\ &= \frac{1}{2} \times 3 \times 2 \\ A_3 &= 3 \text{ cm}^2 \\ A &= A_1 - A_2 - A_3 \\ &= 77 - 9 - 3 \\ A &= 65 \text{ cm}^2 \end{aligned}$$